

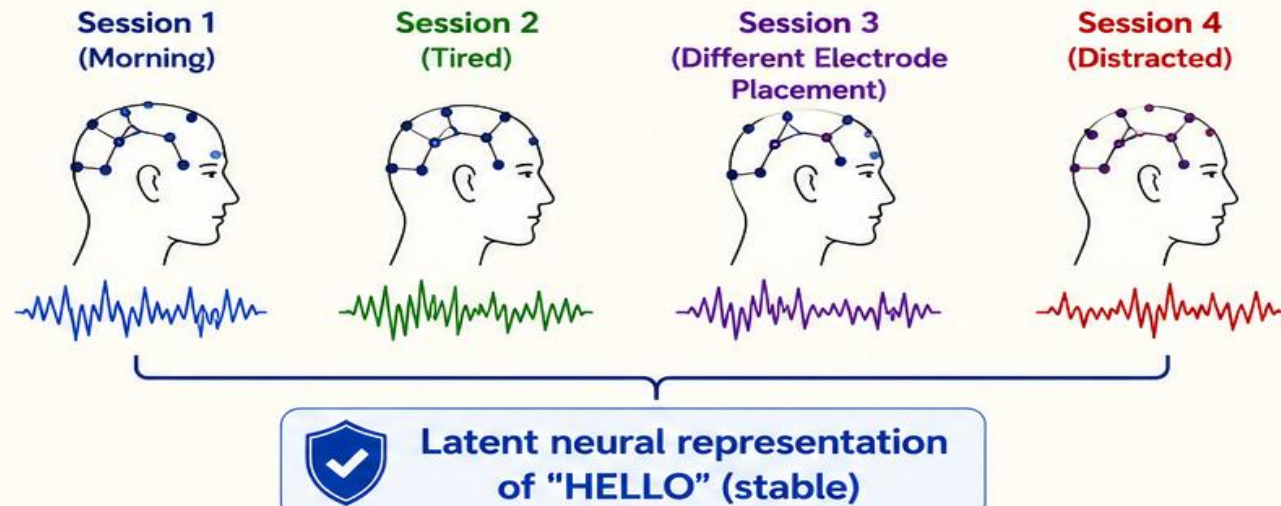


INVARIANT LATENT NEURAL REPRESENTATION LEARNING



1

EEG-based imagined speech decoding suffers from significant **cross-session variability** i.e., no two EEG signals produced in two different sessions are same, which in turn reduces accuracy. Let's take a word 'HELLO'. When you think the word 'HELLO', the raw EEG signal changes every time because of electrode placement, mood, fatigue, attention, noise, and brain state. But perhaps there exists a latent neural representation of 'HELLO' that stays relatively stable. The model's job would be to **learn** that stable representation while **ignoring the variations** and **comparing** it with subject's real-time EEG recording then predicting the word.



2

Think of it as similar to how **facial recognition** works. Your face under sunlight, indoors, smiling, or wearing glasses produces different images, yet a good model recognises the features that define you, not the surroundings.



3



In simple words, it is based on a hypothesis that there is a **latent neural signature or core representation of our thoughts** that remains relatively stable across sessions, while other changes are due to noise, fatigue, mood, attention, electrode placement, and other factors.

4

Think of it as



5



The model's job would be to learn and work with only the **stable component**.

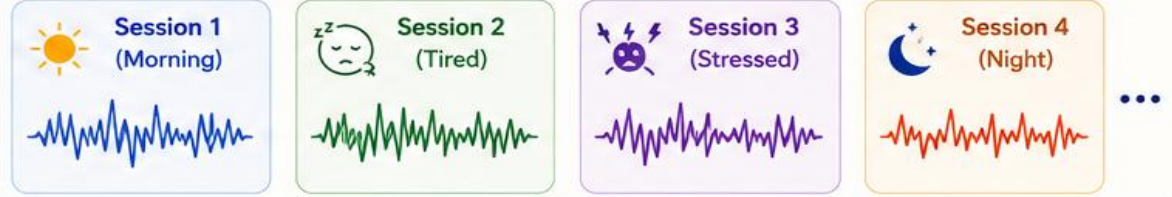


The simplest way to describe the pipeline:

1



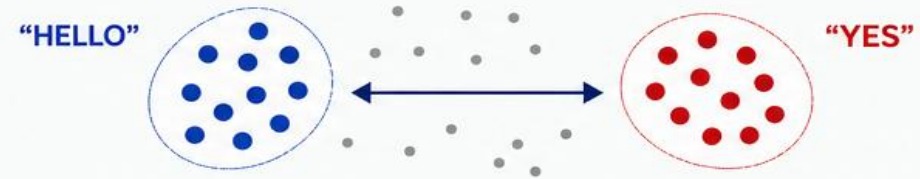
Collect **multiple EEG recordings** of the same word (e.g., "**HELLO**") across different sessions, including different levels of fatigue, attention, stress, and times of day.



2



Train the network to map EEG recordings of the **same word close together** in a shared latent embedding space, while pushing recordings of **different words** (e.g., "**HELLO**" and "**YES**") farther apart.



3



From the clustered embeddings of each word, learn a **prototype (core) representation** that captures the neural characteristics common across all sessions ignoring/suppressing session-specific variations.



4



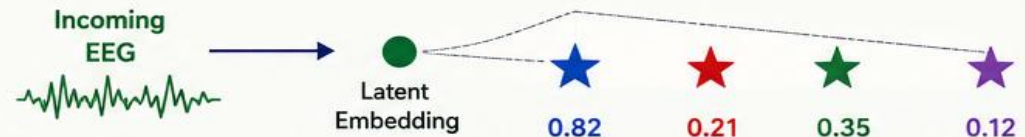
Store these prototype representations in a **memory gallery**, where each prototype serves as the reference representation for its corresponding word.



5



During real-time decoding, transform the incoming EEG signal into the **same latent embedding space** and compare it with the stored prototypes using a similarity metric.



6

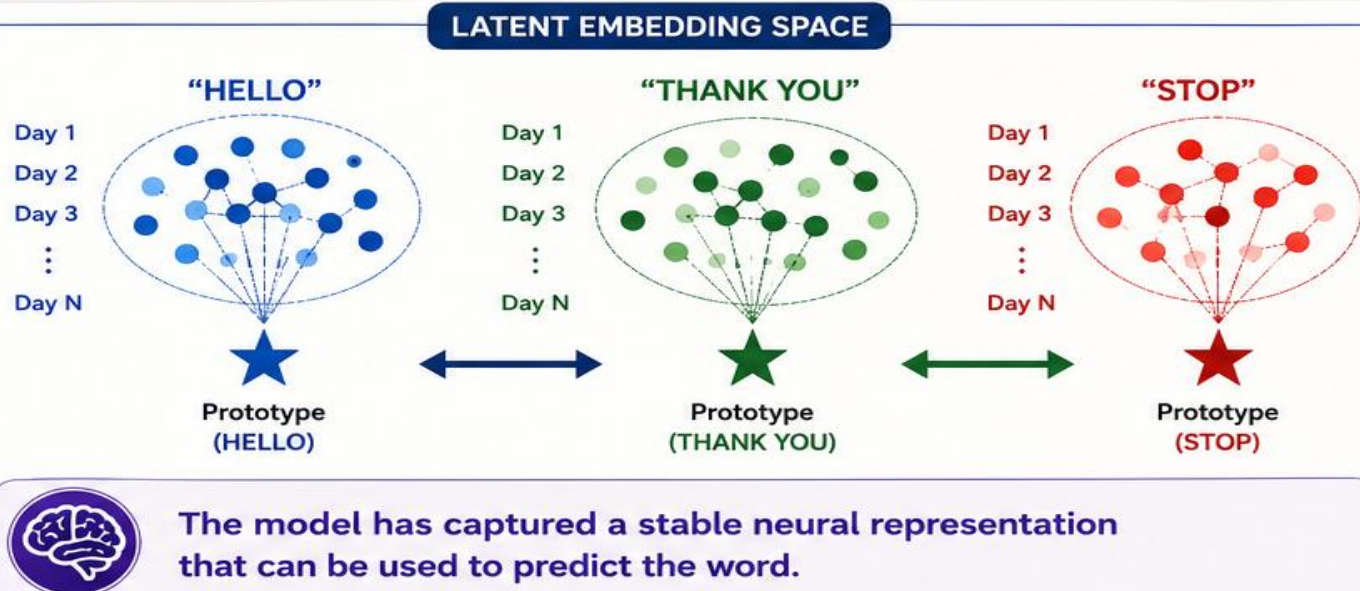


The prototype with the **highest similarity** is selected, and its corresponding word is **predicted**.



“

So if the embeddings for "HELLO" from different days all cluster together while remaining distinct from "THANK YOU" or "STOP" then learn a representative prototype embedding of each word across sessions and we've demonstrated that the model has captured a **stable neural representation** that can be used to predict the word.



THE CHALLENGES



Non-Stationarity:

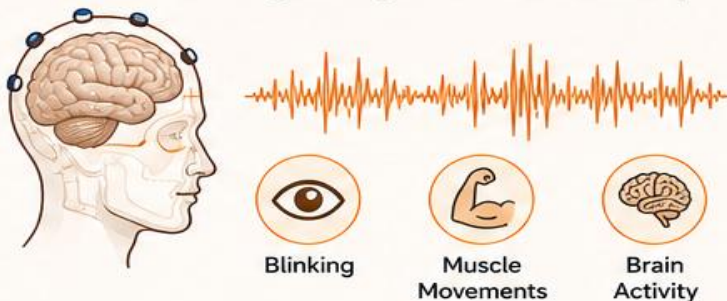
EEG signals shift naturally over hours due to skin sweat, minor headset displacement, fatigue, and baseline electrical drift.

Abstract thought is influenced by memory, emotion, attention, and context, making neural activity less consistent. A channel recording today might look slightly different tomorrow even with zero change in mood.



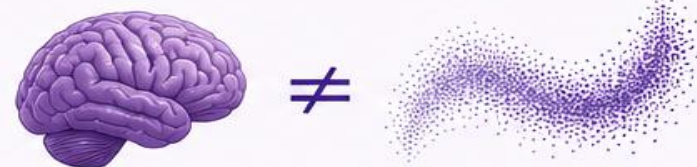
The "Signal-to-Noise" Problem:

EEG sits on the outside of the skull, meaning deep cortical language generation gets heavily blurred by muscle movements, blinking, and general brain activity.



The scientific question is whether such a **stable representation** exists **strongly enough in EEG**.

With invasive recordings, there is good evidence that stable latent neural manifolds can exist but with EEG, the signal is much noisier and lower in spatial resolution, so finding that representation is considerably harder.





FOCUS: ARTICULATORY PHONEMIC DECODING



Since there's a lot of possibility and no absolute certainty of the extraction of a core representation we focus on the **articulatory phonemic decoding**.



This is a shift from **semantic decoding** (what a word 'means') to **articulatory phonemic decoding** (how a word 'feels to pronounce' internally).



It can be said as learning **session-invariant representations** from neural activity associated with phonological encoding and speech motor planning, rather than relying primarily on semantic representations.



It avoids assuming every word has one immutable electrical pattern. Instead, it assumes that the mechanism of internally generating speech sounds is **more stable across sessions** than the raw EEG itself.

While your concept or feeling of "HELLO" changes based on your mood,



Concept / Feeling
(varies with mood)

the mechanical motor plan to imagine saying "H" or "L" uses the **same neural pathways every time**.



"HELLO"



The pronunciation of "HELLO" is almost always the same.



The sequence of phonemes (/h ə l oʊ/) is fixed.



That aligns with how speech is naturally produced by the brain. It's also one reason why many **invasive speech BCIs focus on motor speech cortex**. Those systems decode intended articulatory movements rather than directly decoding abstract meaning.

So this becomes

Instead of:



EEG



Word

We aim for:



EEG



Phoneme sequence
or articulatory
representation



Word

or



EEG



Latent speech
motor representation



Text



It avoids assuming every word has one immutable electrical pattern. Instead, it assumes that the mechanism of internally generating speech sounds is **more stable across sessions** than the raw EEG itself.



The Solution can be through two pathways

1



Decode Covert Speech (Imagined Speech) by targeting the **Broca's area** (motor planning areas) .

It is the neural blueprint to mentally pronounce a syllable and encodes what sounds we intend to sequence together.

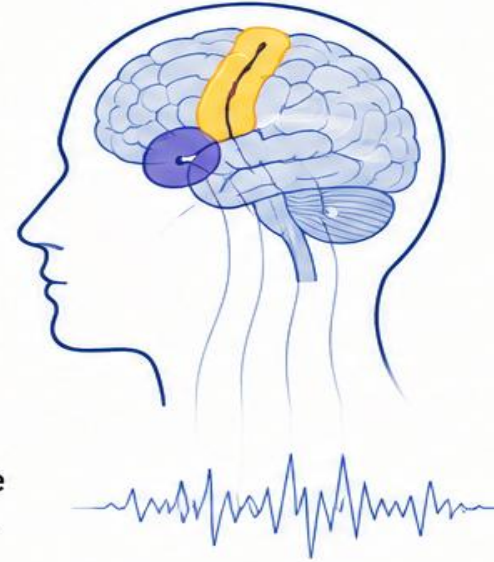
2



Choosing datasets that focus on the speech motor cortex via electrode placement as it encodes how our tongue and lips are mapped to move

Tracking the speech motor cortex is **highly advantageous** as it generates a **more direct, intense, and invariant electrical burst**. When the subject imagines speaking, the speech motor cortex actively flashes the physical structural paths of the tongue and lips, even if they keep their mouth shut.

Broca's area, by contrast, is more abstract, heavily integrated with cognitive networks, and generates quieter, harder-to-isolate signals on scalp EEG.



Primary motivation



Cross-session variability remains one of the biggest limitations reported in the literature. Almost all existing approaches perform well within a session but often fail across different sessions. Frequent recalibration limits practical real-world use.



Our hypothesis can **reduce cross session variability** as every intentionally imagined word is assumed to contain **stable neural characteristics** despite session variability. Learning these **stable prototype representations** can make decoding more robust.

What the model does:

PHASE 1

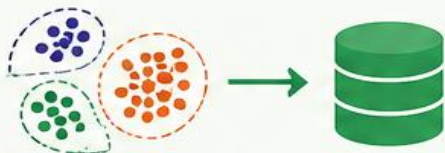
The Training Phase (Session-Invariant Representation Learning)



Instead of training the network as a conventional classifier, the model learns a session-invariant latent embedding space. Multiple EEG recordings of the same imagined word (e.g., "HELLO") are collected across different sessions, representing variations in fatigue, attention, stress, electrode placement, and time of day. The network is trained to map recordings of the same imagined word close together in the latent space while simultaneously pushing recordings of different words farther apart. In doing so, the model suppresses session-specific variability and retains only the neural characteristics that remain consistent for each imagined word.

PHASE 2

The Prototype Learning & Memory Phase (Reference Gallery)



After the latent embedding space has been learned, the clustered embeddings corresponding to each imagined word are used to learn a prototype representation. This prototype represents the common neural characteristics shared across all recordings of that word while minimizing the influence of session-dependent variations. Each learned prototype is stored in a Reference Gallery, where every prototype acts as the permanent reference representation for its corresponding imagined word. Consequently, different imagined words occupy distinct regions of the latent space, enabling clear separation between their neural representations.

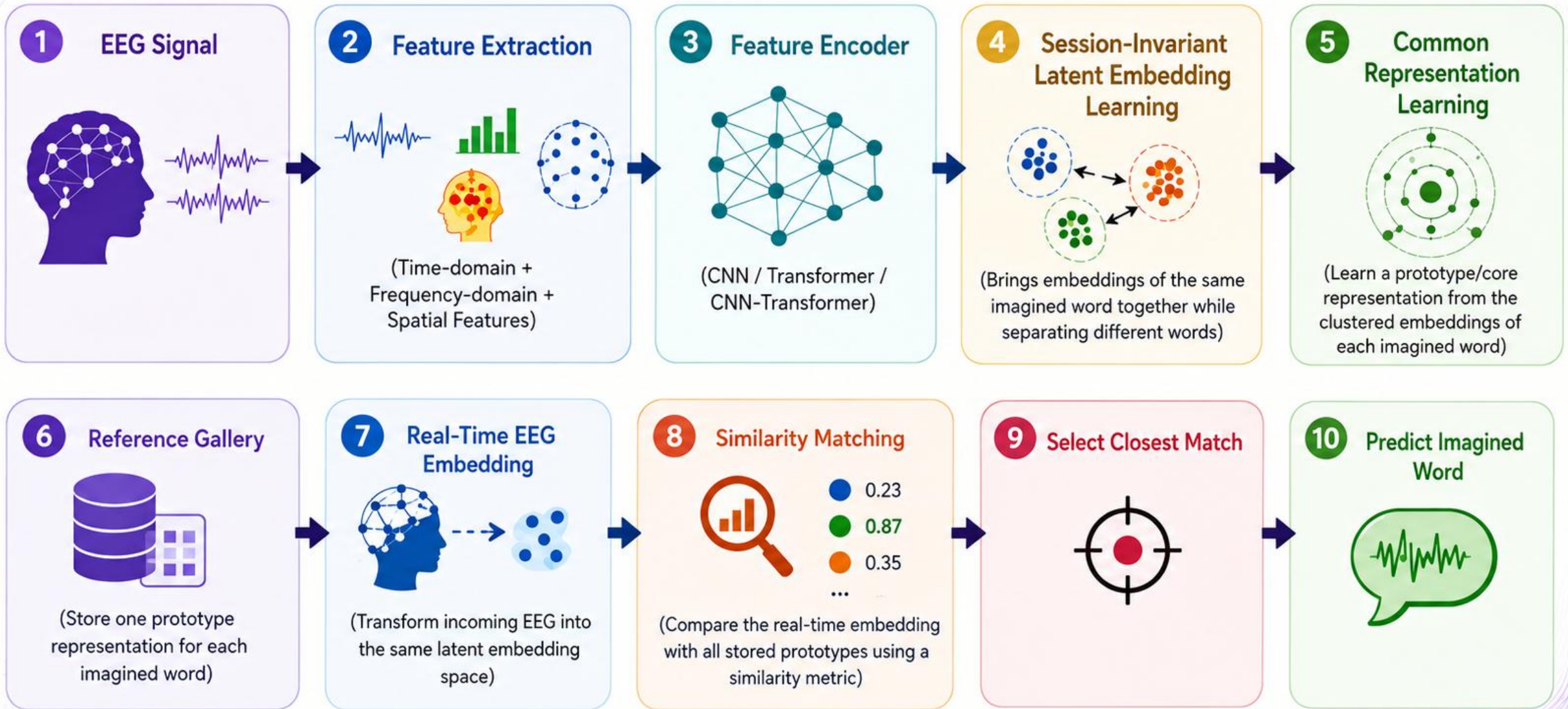
PHASE 3

The Inference Phase (Real-Time Similarity Matching)



During real-time operation, an incoming EEG segment is passed through the trained encoder to transform it into the same latent embedding space. Rather than classifying the signal directly, the system compares the generated embedding with the stored prototypes in the Reference Gallery using a similarity metric. The prototype with the highest similarity score is identified as the closest match, and the imagined word associated with that prototype is predicted. Since only an embedding computation and a rapid similarity search are required during inference, the approach supports low-latency decoding while remaining robust to cross-session variability.

The basic architecture





Subsequent literature analysis identified existing techniques that align with individual components of the proposed framework, reinforcing its feasibility.



These existing techniques can be integrated to optimize the execution phase of our proposed method.



Contrastive learning

A self-supervised technique that teaches models to pull similar data points (positive pairs) closer together and push dissimilar ones (negative pairs) apart in an embedding space.



Metric spaces

Involves defining a mathematical distance function $d(x,y)$ over a set of data points, serving as the foundation for Deep Metric Learning (DML) where neural networks map raw data into structured embedding spaces.



InfoNCE (Information Noise Contrastive Estimation) loss

A contrastive learning objective used in brain-computer interface (BCI) inner speech decoding to align high-dimensional neural representations with semantic or acoustic text embeddings.



Triplet loss

A metric learning objective that trains neural networks to map data into an embedding space where similar items are clustered close together, and dissimilar items are pushed apart.



A Siamese Neural Network

An architecture containing two or more identical sub-networks that share the same weights and configuration to evaluate the similarity between two distinct inputs.



Prototype learning

It classifies data by comparing input features to learned representative examples called prototypes.



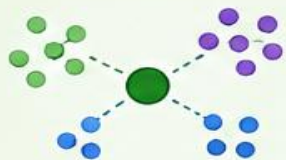
Memory-based networks

Combines traditional neural architectures with external or explicit memory components, allowing models to read from and write to a dynamic storage space rather than relying solely on fixed internal weights.



A prototypical network

A model used for few-shot classification, where it learns a metric space to classify new data points by computing distances to class prototypes.



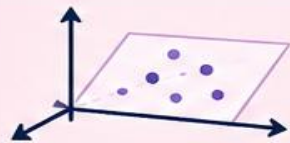
Prototype-guided continual learning

Represents data classes or tasks as compact vectors (prototypes) in an embedding space to prevent catastrophic forgetting.



Disentangled representation learning

Aims to separate distinct, independent, and informative generative factors of variation in data so that individual latent dimensions control specific semantic attributes.



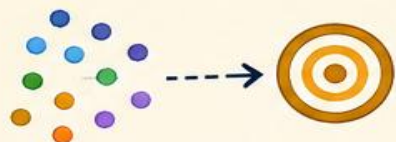
Invariant subspace mapping

Refers to identifying or projecting neural network representations into specific vector spaces that remain unchanged under data transformations, system dynamics, or linear operators.



Session-invariant representation learning

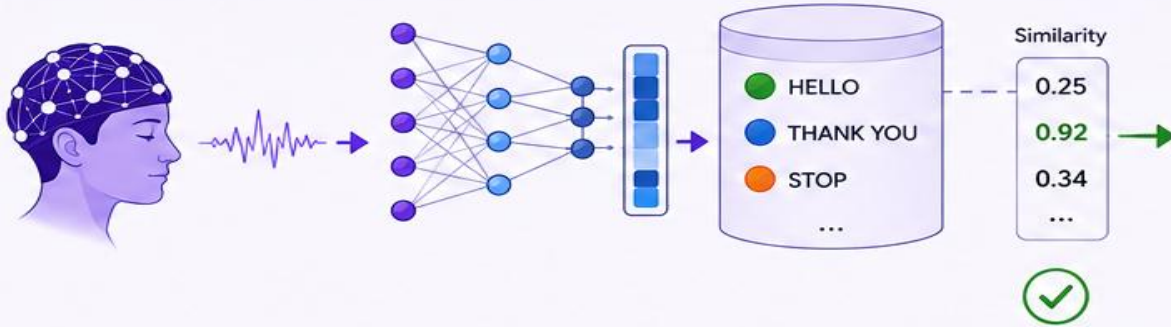
Aims to extract stable, task-relevant features from time-series or multi-session data (such as EEG brain signals or user interaction sessions) while removing session-specific variations, noise, and distribution shifts.



Similarity-based decoding

Refers to computational techniques that use distance or closeness metrics in a representation space to reconstruct or route data

How this framework can **lower latency** ⚡



- During real-time decoding, the incoming EEG signal is converted directly into a latent embedding using the trained encoder.
- Instead of analysing the raw EEG from scratch, the model quickly compares the new embedding with the stored prototype representations in the Reference Gallery using a similarity metric.
- The prototype with the highest similarity score is selected, and its corresponding imagined word is predicted.



Less time is spent handling **session-to-session variability during inference** because the encoder learns session-invariant neural representations instead of adapting to session-specific changes.



The model performs **fast prototype matching** instead of repeatedly analysing complex EEG patterns or performing full classification for every incoming signal.



Most of the computational effort occurs during **offline training**, where session-invariant embeddings and word prototypes are learned, making real-time decoding highly efficient.



The learned prototype representations remain **stable across recording sessions**, reducing the need for frequent recalibration and minimizing retraining for individual sessions.



Limitations of this framework



01



Assumption of a Stable Neural Representation

The framework assumes that every imagined word has a stable latent representation that remains consistent across sessions.

02



Dependence on Large Multi-Session Training Data

The model requires many recordings of the same imagined word under different conditions.

03



Inter-Subject Variability

The method mainly addresses cross-session variability.

04



Prototype Drift Over Time

Brain activity changes over months or years because of aging, learning, fatigue, medication, or long-term physiological changes.

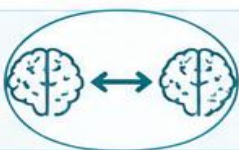
05



Dependence on Quality Feature Extraction

The entire framework depends on the encoder learning meaningful features.

06



Similar Neural Patterns Between Words

Some imagined words may activate highly overlapping neural patterns.

07



Computational Cost During Training

Inference is fast, but training is not.